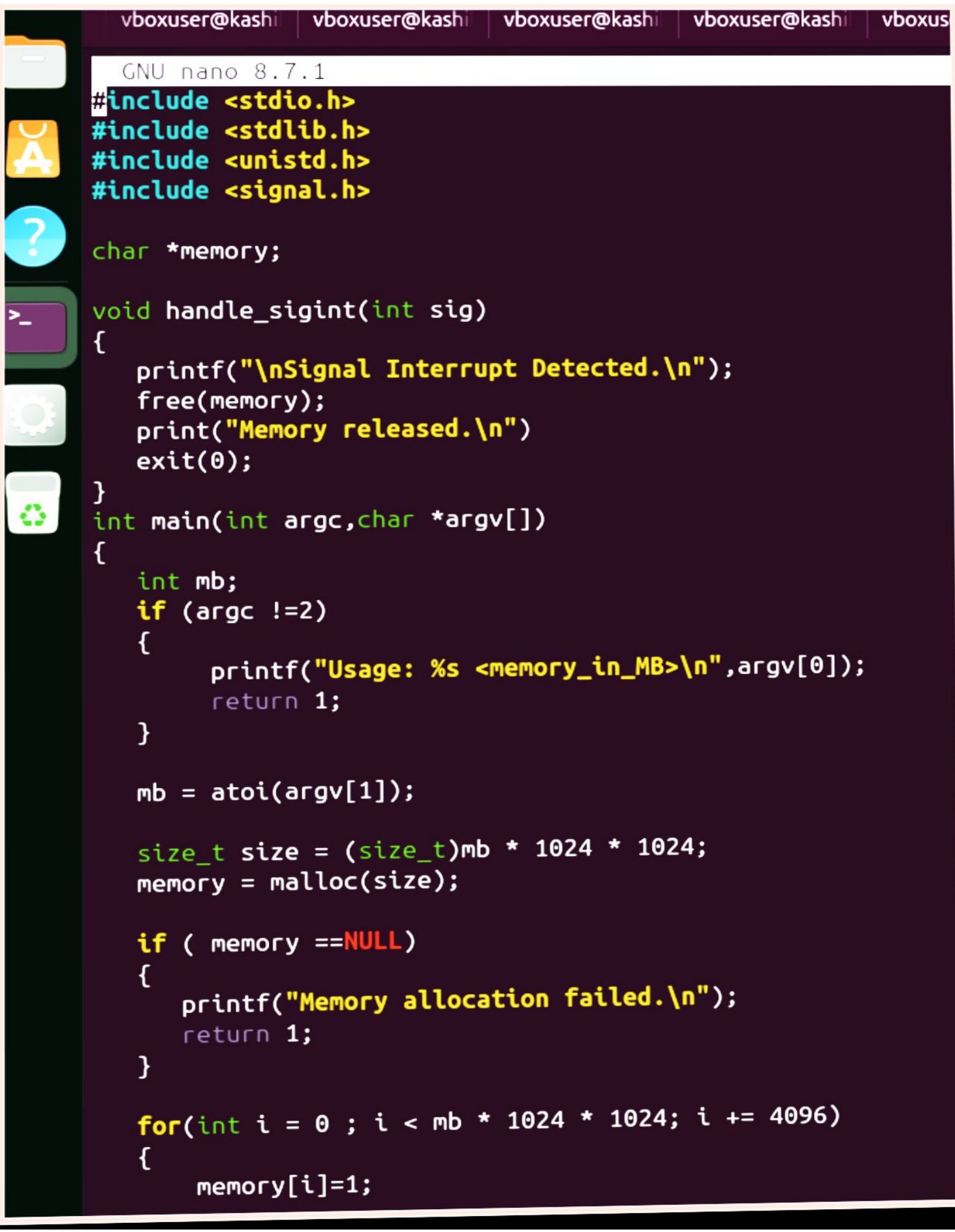

vboxuser@kashi
vboxuser@kashi
vboxuser@kashi
vboxuser@kashi
vboxus
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
char *memory;
void handle_sigint(int sig)
{
printf("\nSignal Interrupt Detected.\n");
free(memory);
print("Memory released.\n")
exit(0);
}
int main(int argc,char *argv[])
{
int mb;
if (argc !=2)
{
printf("Usage: %s <memory_in_MB>\n",argv[0]);
return 1;
}
mb = atoi(argv[1]);
size_t size = (size_t)mb * 1024 * 1024;
memory = malloc(size);
if (memory ==NULL)
{
printf("Memory allocation failed.\n");
return 1;
}
for(int i = 0 ; i < mb * 1024 * 1024; i += 4096)
{
memory[i]=1;

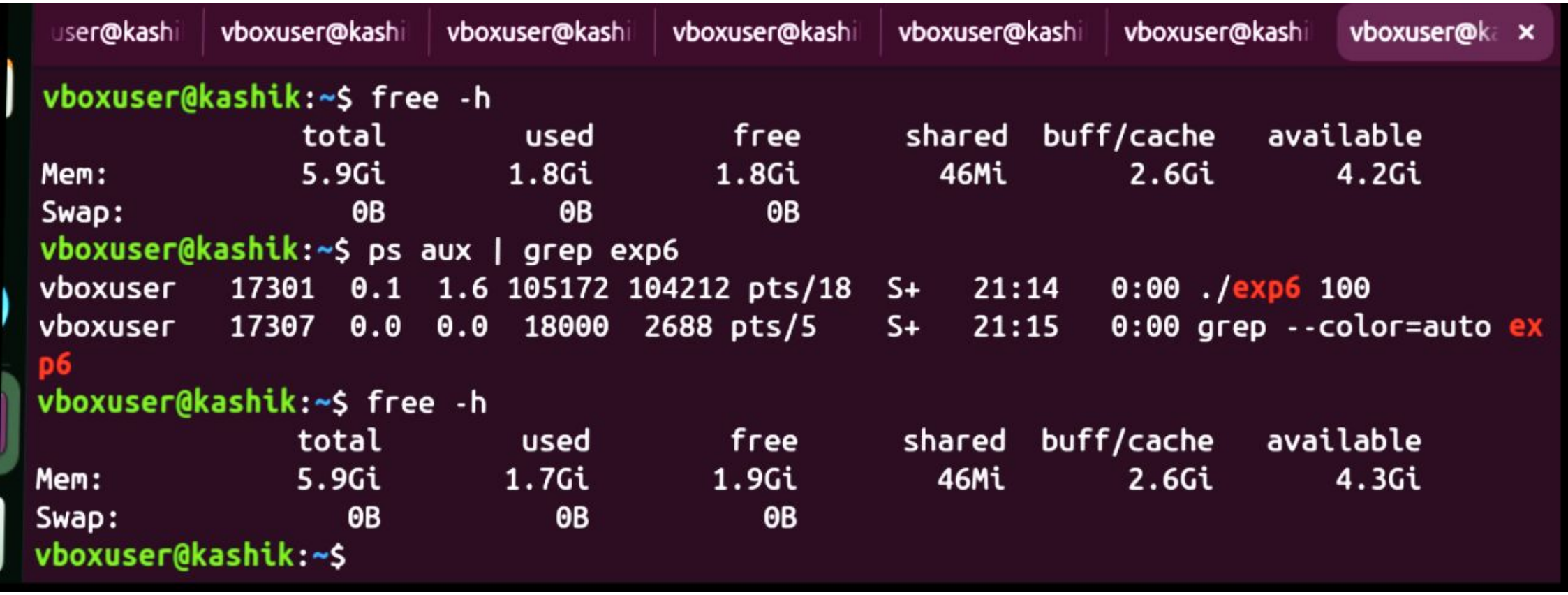
```c
    }

    printf("Memory allocated Successfully.\n");
    printf("Press Ctrl+C to release memory and exit.\");

    signal(SIGINT, handle_sigint);

    while (1)
    {
        sleep(1);
    }

    return  0;
}
```


user@kashi
vboxuser@kashi
vboxuser@kashi
vboxuser@kashi
vboxuser@kashi
vboxuser@kashi
vboxuser@k
vboxuser@kashik:~$ free -h
total used free shared buff/cache available
Mem: 5.9Gi 1.8Gi 1.8Gi 46Mi 2.6Gi 4.2Gi
Swap: 0B 0B 0B
vboxuser@kashik:~$ ps aux | grep exp6
vboxuser 17301 0.1 1.6 105172 104212 pts/18 S+ 21:14 0:00 ./exp6 100
vboxuser 17307 0.0 0.0 18000 2688 pts/5 S+ 21:15 0:00 grep --color=auto exp6
vboxuser@kashik:~$ free -h
total used free shared buff/cache available
Mem: 5.9Gi 1.7Gi 1.9Gi 46Mi 2.6Gi 4.3Gi
Swap: 0B 0B 0B
vboxuser@kashik:~$

```
vboxuser@kashi | vboxuser@kashi | vboxuser@kashi | vbo
vboxuser@kashik:~$ nano exp5.sh
vboxuser@kashik:~$ chmod +x exp5.sh
vboxuser@kashik:~$ ./exp5.sh
Enter a String
Kashik Balla
Length of  String: 12
Enter String Position:
3
Enter Number of character:
5
Substring: hik B
vboxuser@kashik:~$ gcc exp5.c -o string
vboxuser@kashik:~$ ./string
Enter a string:CRISTIANO RONALDO
Length of string: 17
Enter Starting Position: 7
Enter Number of charcter:8
Substring: NO RONAL
vboxuser@kashik:~$ time ./string
Enter a string:Kashik Balla
Length of string: 12
Enter Starting Position: 3
Enter Number of charcter:5
Substring: hik B

real    0m13.404s
user    0m0.001s
sys     0m0.001s
vboxuser@kashik:~$ time ./exp5.sh
Enter a String
Kashik Balla
Length of  String: 12
Enter String Position:
3
Enter Number of character:
5
Substring: hik B

real    0m25.625s
user    0m0.004s
sys     0m0.001s
vboxuser@kashik:~$
```

vboxuser@kashi | vboxuser@kashi | vboxuser@kashi | vboxus

GNU nano 8.7.1

```c
#include <stdio.h>

int main()
{
    char str[100];
    int start,length,i,count=0;

   printf("Enter a string:");
   fgets(str,100,stdin);

   while(str[count] != '\0' && str[count] != '\n')
   {
         count++;
   }

   printf("Length of string: %d\n",count);
   printf("Enter Starting Position: ");
   scanf("%d", &start);
   printf("Enter Number of charcter:");
   scanf("%d", &length);

   printf("Substring: ");

  for (i = start;i<start+length && i < count; i++)
  {
      printf("%c",str[i]);
  }
  printf("\n");

 return 0;
}
```

```
  GNU nano 8.7.1
#!/bin/bash

echo "Enter a String"
read str

echo "Length of  String: ${#str}"

echo "Enter String Position:"
read start

echo "Enter Number of character:"
read length

echo -n "Substring: "
echo "${str:$start:$length}"
```



vboxuser@O

vboxuser@OS1: ~ — nano expt9.c

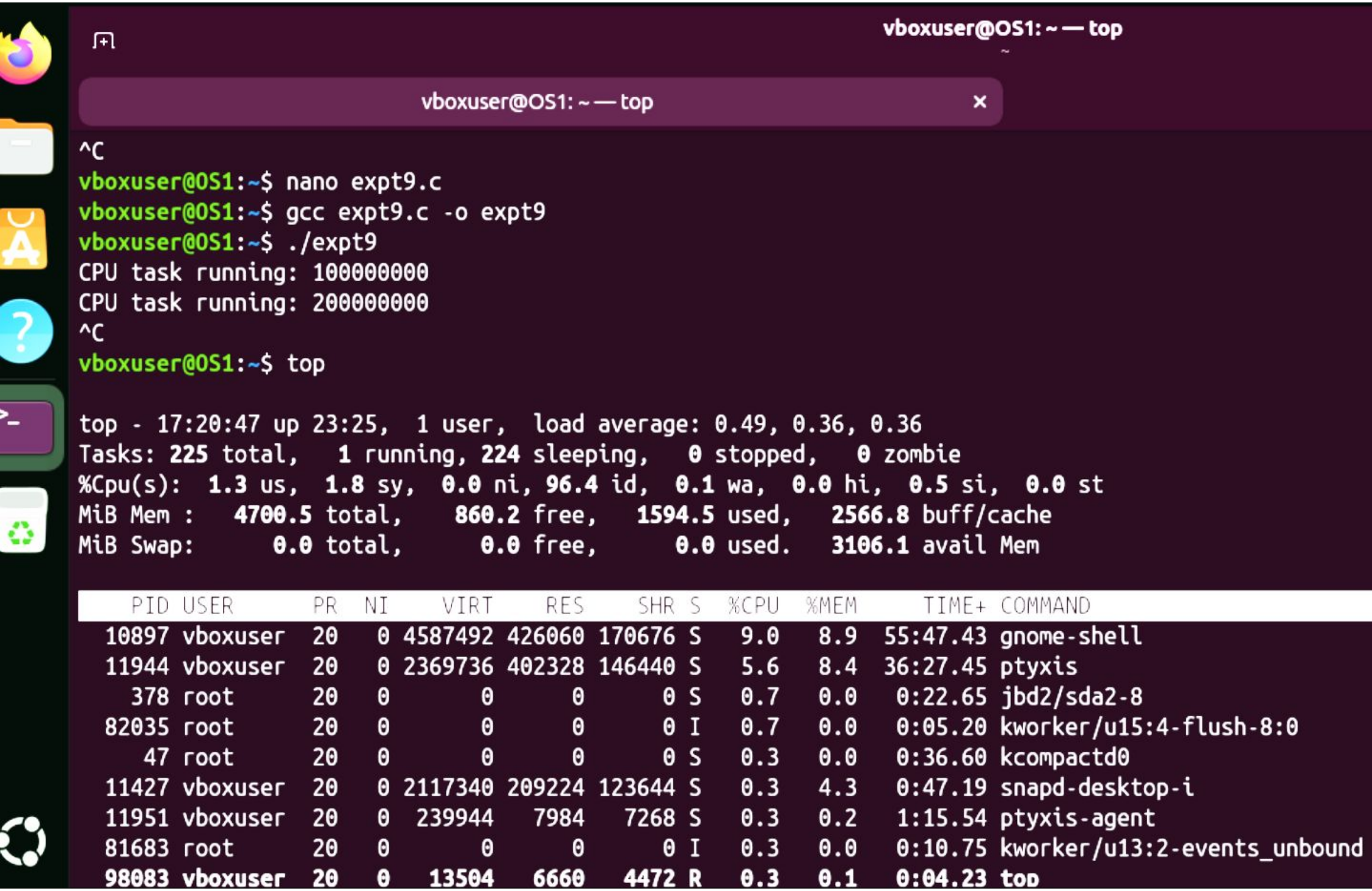
GNU nano 8.7.1

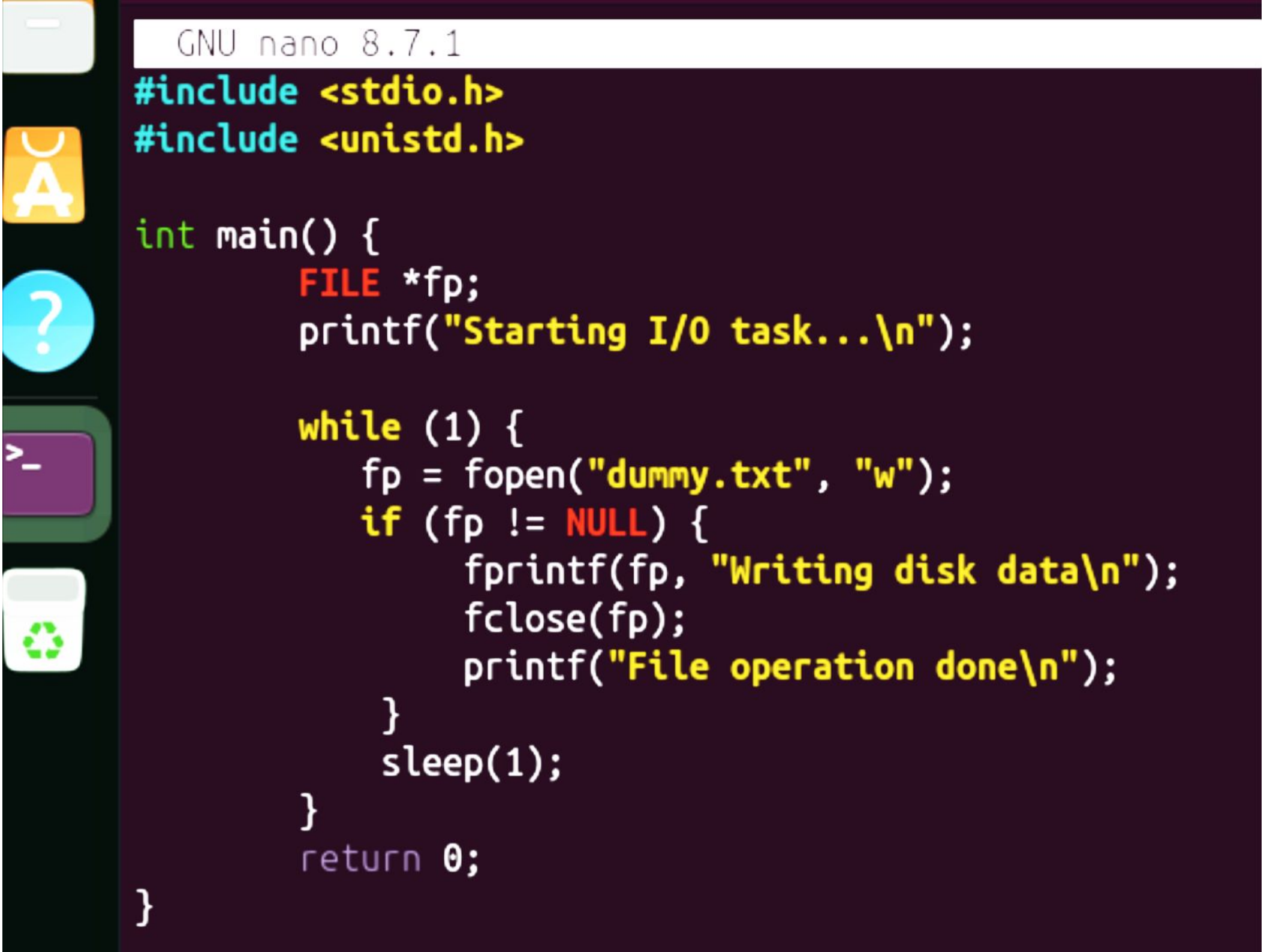
```c
#include <stdio.h>

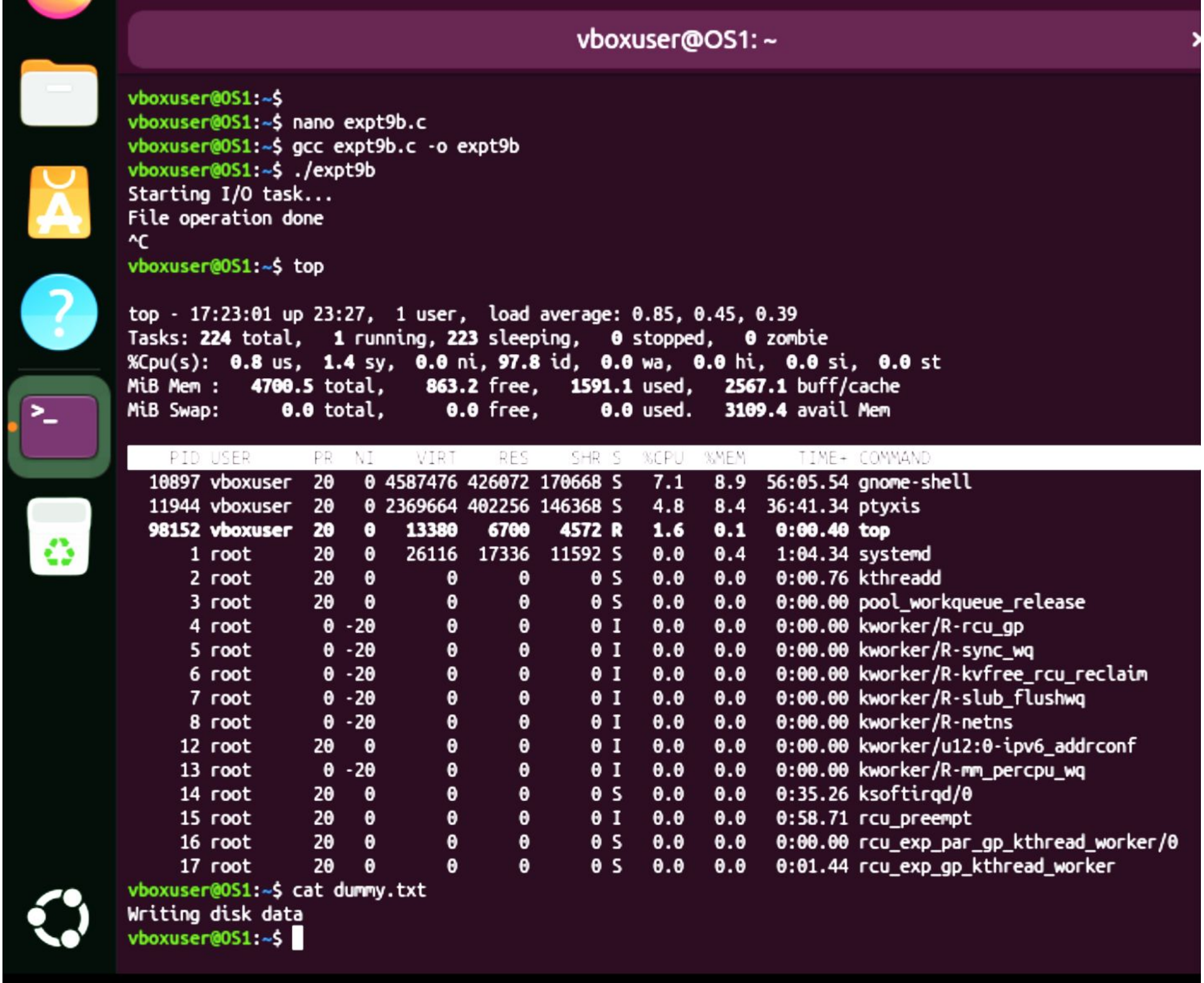
int main() {
    long long counter = 0;

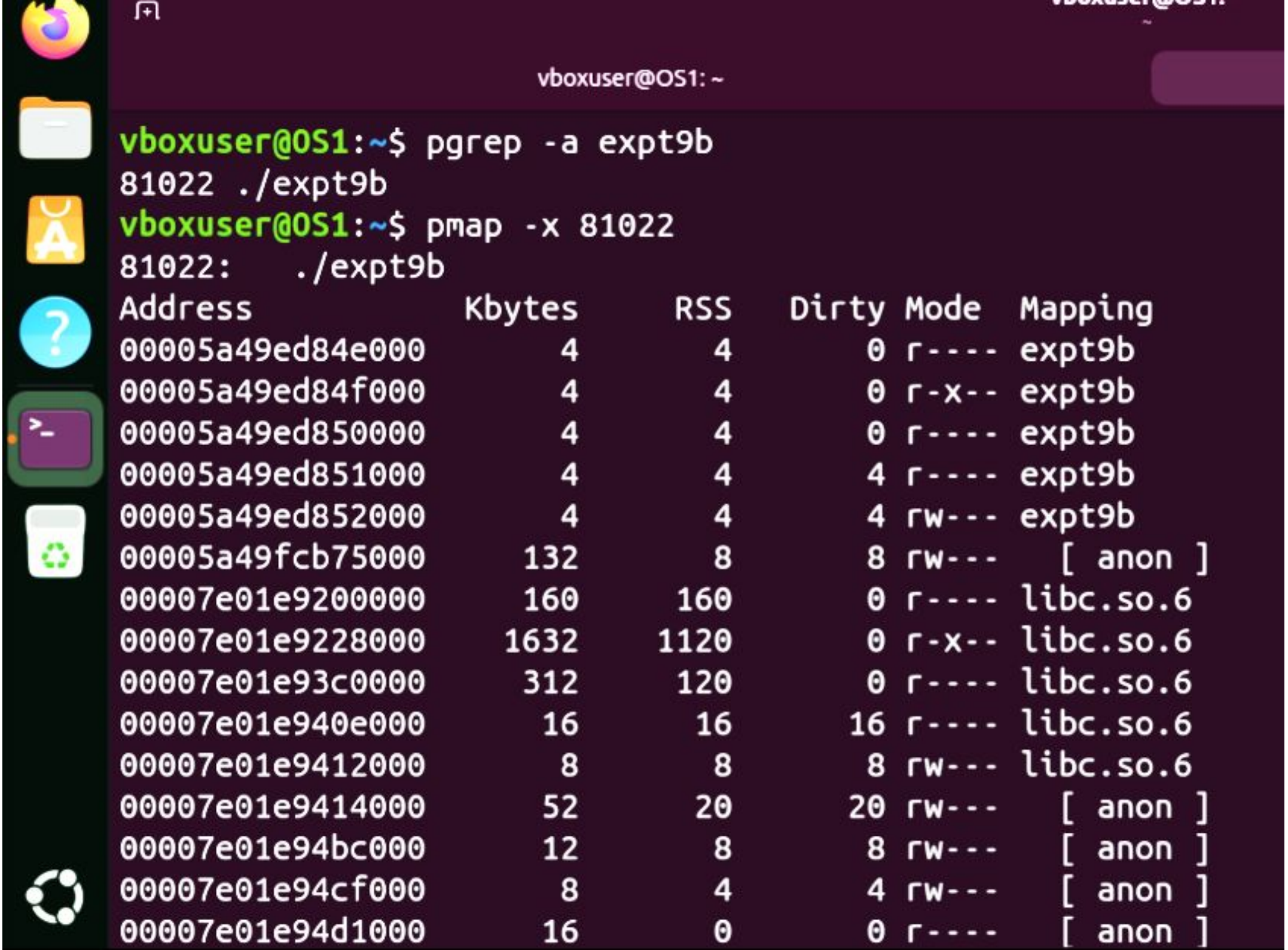
    while (1) {
        counter++;
        if (counter % 100000000 == 0) {
            printf("CPU task running: %lld\n", counter);
        }
    }

    return 0;
}
```


vboxuser@OS1: ~ — top
vboxuser@OS1: ~ — top
^C
vboxuser@OS1:~$ nano expt9.c
vboxuser@OS1:~$ gcc expt9.c -o expt9
vboxuser@OS1:~$ ./expt9
CPU task running: 100000000
CPU task running: 200000000
^C
vboxuser@OS1:~$ top
top - 17:20:47 up 23:25, 1 user, load average: 0.49, 0.36, 0.36
Tasks: 225 total, 1 running, 224 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.3 us, 1.8 sy, 0.0 ni, 96.4 id, 0.1 wa, 0.0 hi, 0.5 si, 0.0 st
MiB Mem : 4700.5 total, 860.2 free, 1594.5 used, 2566.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3106.1 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
10897 vboxuser 20 0 4587492 426060 170676 S 9.0 8.9 55:47.43 gnome-shell
11944 vboxuser 20 0 2369736 402328 146440 S 5.6 8.4 36:27.45 ptyxis
378 root 20 0 0 0 0 S 0.7 0.0 0:22.65 jbd2/sda2-8
82035 root 20 0 0 0 0 I 0.7 0.0 0:05.20 kworker/u15:4-flush-8:0
47 root 20 0 0 0 0 S 0.3 0.0 0:36.60 kcompactd0
11427 vboxuser 20 0 2117340 209224 123644 S 0.3 4.3 0:47.19 snapd-desktop-i
11951 vboxuser 20 0 239944 7984 7268 S 0.3 0.2 1:15.54 ptyxis-agent
81683 root 20 0 0 0 0 I 0.3 0.0 0:10.75 kworker/u13:2-events_unbound
98083 vboxuser 20 0 13504 6660 4472 R 0.3 0.1 0:04.23 top


GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
int main() {
FILE *fp;
printf("Starting I/O task...\n");
while (1) {
fp = fopen("dummy.txt", "w");
if (fp != NULL) {
fprintf(fp, "Writing disk data\n");
fclose(fp);
printf("File operation done\n");
}
sleep(1);
}
return 0;
}


vboxuser@OS1: ~
vboxuser@OS1:~$
vboxuser@OS1:~$ nano expt9b.c
vboxuser@OS1:~$ gcc expt9b.c -o expt9b
vboxuser@OS1:~$ ./expt9b
Starting I/O task...
File operation done
^C
vboxuser@OS1:~$ top
top - 17:23:01 up 23:27, 1 user, load average: 0.85, 0.45, 0.39
Tasks: 224 total, 1 running, 223 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.8 us, 1.4 sy, 0.0 ni, 97.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 4700.5 total, 863.2 free, 1591.1 used, 2567.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3109.4 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
10897 vboxuser 20 0 4587476 426072 170668 S 7.1 8.9 56:05.54 gnome-shell
11944 vboxuser 20 0 2369664 402256 146368 S 4.8 8.4 36:41.34 ptyxis
98152 vboxuser 20 0 13380 6700 4572 R 1.6 0.1 0:00.40 top
1 root 20 0 26116 17336 11592 S 0.0 0.4 1:04.34 systemd
2 root 20 0 0 0 0 S 0.0 0.0 0:00.76 kthreadd
3 root 20 0 0 0 0 S 0.0 0.0 0:00.00 pool_workqueue_release
4 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-rcu_gp
5 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-sync_wq
6 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-kvfree_rcu_reclaim
7 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-slub_flushwq
8 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-netns
12 root 20 0 0 0 0 I 0.0 0.0 0:00.00 kworker/u12:0-ipv6_addrconf
13 root 0 -20 0 0 0 I 0.0 0.0 0:00.00 kworker/R-mm_percpu_wq
14 root 20 0 0 0 0 S 0.0 0.0 0:35.26 ksoftirqd/0
15 root 20 0 0 0 0 I 0.0 0.0 0:58.71 rcu_preempt
16 root 20 0 0 0 0 S 0.0 0.0 0:00.00 rcu_exp_par_gp_kthread_worker/0
17 root 20 0 0 0 0 S 0.0 0.0 0:01.44 rcu_exp_gp_kthread_worker
vboxuser@OS1:~$ cat dummy.txt
Writing disk data
vboxuser@OS1:~$


vboxuser@OS1: ~
vboxuser@OS1:~$ pgrep -a expt9b
81022 ./expt9b
vboxuser@OS1:~$ pmap -x 81022
81022: ./expt9b
Address Kbytes RSS Dirty Mode Mapping
00005a49ed84e000 4 4 0 r---- expt9b
00005a49ed84f000 4 4 0 r-x-- expt9b
00005a49ed850000 4 4 0 r---- expt9b
00005a49ed851000 4 4 4 r---- expt9b
00005a49ed852000 4 4 4 rw--- expt9b
00005a49fcb75000 132 8 8 rw--- [anon]
00007e01e9200000 160 160 0 r---- libc.so.6
00007e01e9228000 1632 1120 0 r-x-- libc.so.6
00007e01e93c0000 312 120 0 r---- libc.so.6
00007e01e940e000 16 16 16 r---- libc.so.6
00007e01e9412000 8 8 8 rw--- libc.so.6
00007e01e9414000 52 20 20 rw--- [anon]
00007e01e94bc000 12 8 8 rw--- [anon]
00007e01e94cf000 8 4 4 rw--- [anon]
00007e01e94d1000 16 0 0 r---- [anon]